Geographic and Seasonal Analysis of Surface Water Quality in India Using CPCB Monitoring Data

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1 Title and Student Details

- **Project Title:** Geographic and Seasonal Analysis of Surface Water Quality in India Using CPCB Monitoring Data
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2 Introduction

2.1 Surface Water Quality and Environmental Context

Surface water in rivers, lakes, reservoirs, and wetlands is essential for drinking water supply, agricultural irrigation, industrial development, and ecological stability. In India, surface water resources face continuous pressure from rapid urbanization, discharge of untreated or partially treated domestic sewage, industrial wastewater outfalls, and agricultural non-point source runoff.

Water quality is evaluated using standardized physicochemical and biological indicator parameters defined by the Central Pollution Control Board (CPCB), Ministry of Environment, Forest and Climate Change (MoEFCC), Government of India:

1. **Dissolved Oxygen (DO):** Reflects the concentration of free molecular oxygen available in the water column (mg/L). It is vital for aquatic life. Under CPCB Class C criteria (drinking water source after conventional treatment and disinfection), DO must remain at or above 4.0 mg/L.
2. **Biochemical Oxygen Demand (BOD):** Measures the amount of dissolved oxygen required by aerobic microorganisms to biologically decompose organic matter over 5 days at 20°C (mg/L). Elevated BOD (> 3.0 mg/L) indicates organic sewage and industrial pollution.

3. **Potential of Hydrogen (pH):** Measures the balance between hydrogen and hydroxyl ions on a standard logarithmic scale (0 to 14). CPCB criteria prescribe an acceptable range between 6.5 and 8.5.
4. **Electrical Conductivity (EC):** Measures the ionic carrying capacity of water ($\mu S/cm$), reflecting the concentration of total dissolved mineral salts.

2.2 Geographic and Seasonal Dynamics in India

India exhibits immense climatic and hydrologic diversity. River basins in the Himalayan north (e.g., Ganga, Brahmaputra) receive perennial snowmelt and monsoon rainfall, while peninsular rivers (e.g., Godavari, Krishna, Cauvery) depend primarily on monsoon precipitation.

Seasonal variations, particularly the Indian Summer Monsoon (June through September), heavily influence water quality: - **Monsoon:** High river discharge induces turbulence, re-aeration, and substantial volumetric dilution of organic pollutants, though high runoff increases sediment and turbidity. - **Pre-Monsoon (Summer):** Elevated ambient water temperatures decrease dissolved oxygen solubility, while reduced streamflow concentrates municipal sewage effluents, worsening organic pollution.

2.3 Data Sources and Rational Architecture

This project investigates surface water quality using two distinct, complementary data sources:

1. **Primary Dataset: Official CPCB / National Water Data Portal (NWDP) Data:**
    - Disseminated through the National Water Informatics Centre (NWIC), Ministry of Jal Shakti.
    - Contains **315 state-wise CSV files** comprising **311,379 observations** categorized across Physical, Chemical, and Biological monitoring parameter groups.
    - Spans the modern regulatory monitoring period (**2018–2024**, with core coverage across 2018–2021).
    - Features complete geospatial coordinates (WGS84 decimal latitude and longitude) across **4,017 monitoring stations** across all 36 States and Union Territories of India.
2. **Supporting Historical Dataset: Kaggle Indian Water Quality Data:**
    - Sourced from Kaggle (`anbarivan/indian-water-quality-data`).
    - Comprises **1,991 raw records** (1,990 unique records after duplicate removal) spanning **2003–2014** (12 years of annualized station summary means).
    - Serves as a macro-level historical baseline for state-level temporal context.

Scientific Data Preprocessing Principle: The two datasets have zero temporal overlap (2003–2014 vs. 2018–2024) and represent different aggregation levels (annualized station averages vs. high-frequency periodic event sampling). Therefore, following sound data science methodology, the datasets are **analyzed separately** and compared only via valid state-level aggregations.

---

3 Objectives

The core objectives of this data science investigation are:

1. **Dataset Inspection:** Empirically inspect the raw local file structures, schemas, encoding types, and storage formats across the 315 CPCB files and the Kaggle dataset.

2. **Data Cleaning & Standardization:** Implement transparent, reproducible data cleaning steps: standardizing column names, repairing corrupted headers, harmonizing state names, and converting date timestamps.
3. **Anomaly & Missingness Handling:** Identify missing-value patterns, check duplicate records, distinguish legitimate extreme values from measurement errors, and handle non-physical observations (such as pH > 14 or non-physical DO readings) using documented scientific rules.
4. **Distributional Analysis:** Compute parametric and non-parametric summary statistics (mean, standard deviation, median, IQR) and visualize distributions for DO, BOD, pH, and Electrical Conductivity.
5. **Geographic & Spatial Exploration:** Analyze state-level water quality variations, compare distributions using boxplots, and map the spatial configuration of monitoring stations.
6. **Seasonal Dynamic Evaluation:** Classify observations into meteorological seasons (Pre-Monsoon, Monsoon, Post-Monsoon, Winter) and evaluate seasonal variations in DO and BOD using non-parametric hypothesis tests.
7. **Bivariate & Multivariate Relationships:** Evaluate non-linear dependencies between BOD and DO, and compute a rank correlation matrix across key physicochemical indicators.
8. **Historical Context Comparison:** Compare modern CPCB state-level median DO and BOD with historical baselines (2003–2014) to assess long-term shifts.
9. **Basic Machine Learning (Linear Regression):** Train and evaluate a beginner-level supervised Linear Regression model (`lm()`) using an 80:20 train-test split to predict Dissolved Oxygen (DO) from Biochemical Oxygen Demand (BOD).

---

4 Dataset Description

4.1 Inventory and Empirical Dimensions

We compute dataset dimensions directly from the raw and processed files on disk without hard-coding counts:

```
# Inspect file counts and sizes directly from raw directories
phys_files <- list.files(file.path(RAW_DIR, "nwdp_cpcb", "physical"), pattern = "[.]csv$", full.names =
chem_files <- list.files(file.path(RAW_DIR, "nwdp_cpcb", "chemical"), pattern = "[.]csv$", full.names =
bio_files  <- list.files(file.path(RAW_DIR, "nwdp_cpcb", "biological"), pattern = "[.]csv$", full.names

# Load consolidated domain tables and datasets to compute exact row and column dimensions
df_phys_raw   <- read_csv(file.path(PROC_DIR, "nwdp_cpcb_physical_clean.csv"), show_col_types = FALSE, p
df_chem_raw   <- read_csv(file.path(PROC_DIR, "nwdp_cpcb_chemical_clean.csv"), show_col_types = FALSE, p
df_bio_raw    <- read_csv(file.path(PROC_DIR, "nwdp_cpcb_biological_clean.csv"), show_col_types = FALSE
df_kaggle_raw <- read.csv(file.path(RAW_DIR, "kaggle", "water_dataX.csv"),
                          stringsAsFactors = FALSE, fileEncoding = "latin1", check.names = FALSE)
df_cpcb_master <- read_csv(file.path(PROC_DIR, "nwdp_cpcb_master_water_quality.csv"),
                           show_col_types = FALSE, progress = FALSE)

# Inspect raw file header column counts
get_raw_col_count <- function(f) {
  con <- file(f, "r")
  hdr <- readLines(con, n = 1, warn = FALSE)
  close(con)
  length(strsplit(hdr, ",")[[1]])
}
```

```r
# Empirical Summary Table dynamically calculated from files
ds_summary <- data.frame(
  Dataset = c(
    "CPCB Physical Parameters (Raw)",
    "CPCB Chemical Parameters (Raw)",
    "CPCB Biological Parameters (Raw)",
    "CPCB Master Harmonized (Cleaned)",
    "Kaggle Historical Data (Raw)"
  ),
  Source = c(
    "NWDP / CPCB (Govt of India)",
    "NWDP / CPCB (Govt of India)",
    "NWDP / CPCB (Govt of India)",
    "NWDP / CPCB Harmonized",
    "Kaggle (anbarivan)"
  ),
  Files = c(length(phys_files), length(chem_files), length(bio_files), 1, 1),
  Observations = c(nrow(df_phys_raw), nrow(df_chem_raw), nrow(df_bio_raw), nrow(df_cpcb_master), nrow(d
  Variables = c(get_raw_col_count(phys_files[1]), get_raw_col_count(chem_files[1]), get_raw_col_count(b
  Date_Coverage = c(
    paste(min(df_phys_raw$year, na.rm = TRUE), max(df_phys_raw$year, na.rm = TRUE), sep = "-"),
    paste(min(df_chem_raw$year, na.rm = TRUE), max(df_chem_raw$year, na.rm = TRUE), sep = "-"),
    paste(min(df_bio_raw$year,  na.rm = TRUE), max(df_bio_raw$year,  na.rm = TRUE), sep = "-"),
    paste(min(df_cpcb_master$year, na.rm = TRUE), max(df_cpcb_master$year, na.rm = TRUE), sep = "-"),
    paste(min(as.numeric(df_kaggle_raw$year), na.rm = TRUE), max(as.numeric(df_kaggle_raw$year), na.rm =
  ),
  Spatial_Coords = c("WGS84 Lat/Lon", "WGS84 Lat/Lon", "WGS84 Lat/Lon", "100% Lat/Lon", "Absent (0%)"),
  stringsAsFactors = FALSE
)

kable(ds_summary, caption = "Table 1: Empirical Dataset Summary Computed Directly from Local Files",
      format.args = list(big.mark = ","))
```

Table 1: Table 1: Empirical Dataset Summary Computed Directly from Local Files

Dataset	Source	Files	Observations	Variables	Date_Coverage	Spatial_Coords
CPCB Physical Parameters (Raw)	NWDP / CPCB (Govt of India)	104	94,159	25	2018-2021	WGS84 Lat/Lon
CPCB Chemical Parameters (Raw)	NWDP / CPCB (Govt of India)	106	109,683	52	2018-2024	WGS84 Lat/Lon
CPCB Biological Parameters (Raw)	NWDP / CPCB (Govt of India)	105	107,537	23	2018-2021	WGS84 Lat/Lon
CPCB Master Harmonized (Cleaned)	NWDP / CPCB Harmonized	1	109,422	22	2018-2024	100% Lat/Lon
Kaggle Historical Data (Raw)	Kaggle (anbarivan)	1	1,991	12	2003-2014	Absent (0%)

4.2 Schema Comparison and Domain Harmonization

The official CPCB dataset separates observations across three parameter categories with distinct schemas:
- **Physical Parameters (104 files, 25 columns):** Contains 19 station/location metadata columns

and 6 physical parameters: `Electric Conductivity (uS/cm)`, `Colour`, `Odour`, `Turbidity (NTU)`, `Temperature (C)`, and `Total Solids (mg/L)`. - **Chemical Parameters (106 files, 52 columns in 105 files):** Contains 19 station metadata columns and 33 chemical parameters including `Dissolved oxygen (mg/L)`, `Potential of Hydrogen (pH)`, `Amonia N (mgN/L)`, `Nitrate N (mgN/L)`, and trace heavy metals (`Arsenic`, `Lead`, `Cadmium`, `Iron`, `Mercury`). *Schema Audit Finding:* One file in the chemical folder (`swq_manual_biological_parameters_cwc_ml_2021_2025.csv`) actually follows the 23-column biological schema and was misclassified in the source archive. - **Biological Parameters (105 files, 23 columns):** Contains 19 station metadata columns and 4 biological parameters: `Biochemical Oxygen Demand (mg/L)`, `Chemical Oxygen Demand (mg/L)`, `Fecal Coliform (MPN/100mL)`, and `Total Coliform (MPN/100mL)`.

Important Note on Domain Harmonization: Because the physical (94,159 rows), chemical (109,683 rows), and biological (107,537 rows) tables have different row counts and schemas, they are **never combined by row position**. Instead, `nwdp_cpcb_master_water_quality.csv` (109,422 rows) was constructed by relational joining on the composite key `(station, data_acquisition_time)`, aligning measurements recorded at the exact same station and timestamp across 22 harmonized variables:

```
# Variable classification in CPCB Master
num_vars  <- names(df_cpcb_master)[sapply(df_cpcb_master, is.numeric)]
char_vars <- names(df_cpcb_master)[sapply(df_cpcb_master, is.character)]

cat("Numerical Variables (", length(num_vars), "):\n", paste(num_vars, collapse = ", "), "\n\n")
```

```
## Numerical Variables ( 16 ):
##  latitude, longitude, year, month, ph, dissolved_oxygen_mg_l, ammonia_n_mg_l, nitrate_n_mg_l, arseni
```

```
cat("Categorical / Metadata Variables (", length(char_vars), "):\n", paste(char_vars, collapse = ", "),
```

```
## Categorical / Metadata Variables ( 5 ):
##  station, data_acquisition_time, state, district, season
```

4.3 Missing Value Audit

Missingness varies across parameters in both datasets due to differing monitoring schedules and laboratory protocols:

```
calc_missing_table <- function(df) {
  data.frame(
    Variable = colnames(df),
    Valid_Records = colSums(!is.na(df)),
    Missing_Records = colSums(is.na(df)),
    Missing_Percentage = round(colMeans(is.na(df)) * 100, 2),
    stringsAsFactors = FALSE
  ) %>% arrange(desc(Missing_Percentage))
}

cpcb_miss <- calc_missing_table(df_cpcb_master)
kable(head(cpcb_miss, 12), caption = "Table 2a: Missing Value Audit in Official CPCB Master Dataset")
```

Table 2: Table 2a: Missing Value Audit in Official CPCB Master Dataset

	Variable	Valid_Records	Missing_Records	Missing_Percentage
arsenic_mg_l	arsenic_mg_l	1157	108265	98.94
lead_mg_l	lead_mg_l	1307	108115	98.81
iron_mg_l	iron_mg_l	5332	104090	95.13
nitrate_n_mg_l	nitrate_n_mg_l	25035	84387	77.12
turbidity_ntu	turbidity_ntu	25631	83791	76.58
ammonia_n_mg_l	ammonia_n_mg_l	26618	82804	75.67
total_coliform_mpn_100ml	total_coliform_mpn_100ml	41771	67651	61.83
electrical_conductivity_us_cm	electrical_conductivity_us_cm	48160	61262	55.99
fecal_coliform_mpn_100ml	fecal_coliform_mpn_100ml	73617	35805	32.72
bod_mg_l	bod_mg_l	84272	25150	22.98
dissolved_oxygen_mg_l	dissolved_oxygen_mg_l	96978	12444	11.37
ph	ph	107342	2080	1.90

```
df_kag_clean <- read.csv(file.path(PROC_DIR, "kaggle_water_quality_clean.csv"), stringsAsFactors = FALSE
kag_miss <- calc_missing_table(df_kag_clean)
kable(kag_miss, caption = "Table 2b: Missing Value Audit in Cleaned Kaggle Historical Dataset")
```

Table 3: Table 2b: Missing Value Audit in Cleaned Kaggle Historical Dataset

	Variable	Valid_Records	Missing_Records	Missing_Percentage
fecal_coliform_mpn_100ml	fecal_coliform_mpn_100ml	1675	315	15.83
nitrate_nitrite_n_mg_l	nitrate_nitrite_n_mg_l	1766	224	11.26
location	location	1807	183	9.20
total_coliform_mpn_100ml_mean	total_coliform_mpn_100ml_mean	1859	131	6.58
station_code	station_code	1869	121	6.08
ph	ph	1894	96	4.82
temperature_c	temperature_c	1899	91	4.57
bod_mg_l	bod_mg_l	1948	42	2.11
dissolved_oxygen_mg_l	dissolved_oxygen_mg_l	1960	30	1.51
conductivity_umhos_cm	conductivity_umhos_cm	1966	24	1.21
state	state	1983	7	0.35
year	year	1990	0	0.00

```
knitr::include_graphics(file.path(FIG_DIR, "01_missingness_overview.png"))
```

Interpretation of Missingness Overview: In both datasets, core regulatory parameters (`ph`, `dissolved_oxygen_mg_l`, and `bod_mg_l`) exhibit high completeness ($> 77\%$ valid in CPCB; $> 95\%$ valid in Kaggle). Conversely, trace heavy metals and specialized bacteriological counts have higher missingness rates in CPCB because they are sampled at lower frequencies. We retain `NA` values rather than applying mean or median imputation, as artificial imputation would distort natural spatial and seasonal variance.

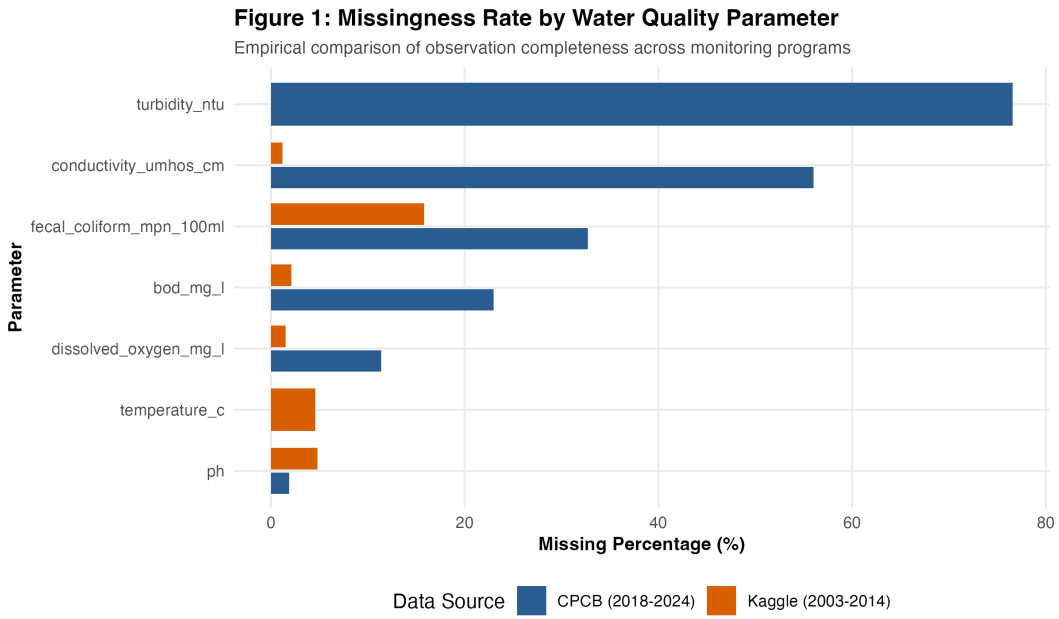


Figure 1: Figure 1: Missingness Rate by Water Quality Parameter Across Datasets

5 Data Cleaning and Preprocessing

Data preprocessing is a pivotal phase of the data science lifecycle. Real-world regulatory and scraped datasets contain missing values, non-standard text markers, encoding anomalies, and transcription errors.

5.1 Step 1: Raw Data Ingestion & Encoding Resolution

The Kaggle dataset contains a micro symbol in `CONDUCTIVITY (µmhos/cm)` that fails under standard UTF-8 parsing in base R (`invalid multibyte string`). Specifying `fileEncoding = "latin1"` cleanly resolves this issue without data loss.

```
# Ingest raw Kaggle dataset with latin1 encoding
df_k_step1 <- read.csv(file.path(RAW_DIR, "kaggle", "water_dataX.csv"),
                       stringsAsFactors = FALSE, fileEncoding = "latin1", check.names = FALSE)
cat("Kaggle successfully ingested. Initial dimensions:", nrow(df_k_step1), "rows,", ncol(df_k_step1), "
```

```
## Kaggle successfully ingested. Initial dimensions: 1991 rows, 12 columns.
```

Academic Interpretation: When importing external CSV files generated across different operating systems, encoding mismatches must be diagnosed early before applying automated parsers.

5.2 Step 2: Column Header Standardization

Raw column headers in both sources contain mixed casing, special characters, spaces, and OCR artifacts (e.g., `NITRATENAN N+ NITRITENANN (mg/l)`). We convert them into standard lowercase `snake_case`.

```r
new_names <- c(
  "station_code", "location", "state", "temperature_c",
  "dissolved_oxygen_mg_l", "ph", "conductivity_umhos_cm",
  "bod_mg_l", "nitrate_nitrite_n_mg_l", "fecal_coliform_mpn_100ml",
  "total_coliform_mpn_100ml_mean", "year"
)
colnames(df_k_step1) <- new_names
cat("Column names standardized to snake_case. Repaired OCR defect in nitrate column.\n")
```

```
## Column names standardized to snake_case. Repaired OCR defect in nitrate column.
```

5.3 Step 3: Duplicate Record Identification and Elimination

Duplicate records inflate sample size and bias statistical estimates:

```r
dups_count <- sum(duplicated(df_k_step1))
cat("Identified full-row duplicates in Kaggle data:", dups_count, "\n")
```

```
## Identified full-row duplicates in Kaggle data: 1
```

```r
df_k_step1 <- df_k_step1[!duplicated(df_k_step1), ]
cat("Records remaining post duplicate elimination:", nrow(df_k_step1), "\n")
```

```
## Records remaining post duplicate elimination: 1990
```

5.4 Step 4: Missingness Marker Normalization and Numeric Conversion

Missing values in the Kaggle file are encoded as `"NAN"`, `""`, `"NA"`, or `"-"`, causing numeric columns to be imported as character strings. We systematically convert these markers to standard R `NA` and cast parameters to numeric.

```r
num_cols <- c("temperature_c", "dissolved_oxygen_mg_l", "ph", "conductivity_umhos_cm",
              "bod_mg_l", "nitrate_nitrite_n_mg_l", "fecal_coliform_mpn_100ml",
              "total_coliform_mpn_100ml_mean", "year")

for (col in num_cols) {
  raw_v <- df_k_step1[[col]]
  df_k_step1[[col]] <- suppressWarnings(as.numeric(ifelse(raw_v %in% c("NAN", "NA", "", "-", " "), NA, r
}
cat("Numeric columns successfully converted. Data types verified.\n")
```

```
## Numeric columns successfully converted. Data types verified.
```

5.5 Step 5: Range Validation, Zero-Value Assessment & Outlier Filtering

A critical tenet of environmental data science is distinguishing **physically impossible measurement errors** from **legitimate extreme pollution values** and **genuine zero measurements**:

1. **Potential of Hydrogen (pH):** Defined strictly on the logarithmic scale $[0, 14]$. Values above 14 represent PDF scraping column-shift artifacts (e.g., conductivity values like 239 or 67,115 shifted into the pH column) or decimal omissions (e.g., 78.2 instead of 7.82). In the raw Kaggle dataset, **89 records** exhibited pH > 14; in the raw CPCB chemical dataset, **4 records** exhibited pH > 14 (up to 78.2). Rather than dropping these rows entirely, we set only the invalid pH values to `NA`, preserving valid measurements for DO, BOD, and conductivity on those rows.
2. **Dissolved Oxygen (DO):** Normal aqueous saturation at 20°C is approximately 9.1 mg/L, and supersaturation in natural waters rarely exceeds 20–25 mg/L. In the raw CPCB chemical data, **11 records** showed physically impossible DO values exceeding 25 mg/L (up to 77.0 mg/L). These 11 values were converted to `NA`.
3. **Zero-Value Assessment (`DO == 0` vs. `EC <= 0`):** In the CPCB dataset, **98 observations** record DO $= 0.0$ mg/L. In environmental chemistry, DO $= 0$ mg/L is a **genuine physical measurement** representing complete anoxia (septic conditions) in heavily polluted river stretches; therefore, zero DO values are retained. Conversely, EC ≤ 0 μS/cm is physically impossible in natural surface water.
4. **Retention of Legitimate Extreme Values (BOD and EC):** BOD values up to 1,315 mg/L and Electrical Conductivity values up to 25,392,600 μS/cm occur in CPCB records near industrial outfalls, sewage drains, and estuarine/tidal zones. Because these represent real extreme environmental conditions rather than impossible bounds, they are retained in the dataset and analyzed using robust non-parametric statistics (median, IQR, rank tests) and logarithmic transformations.

```
# Kaggle pH cleaning
bad_ph_k <- sum(!is.na(df_k_step1$ph) & (df_k_step1$ph < 0 | df_k_step1$ph > 14))
df_k_step1$ph[!is.na(df_k_step1$ph) & (df_k_step1$ph < 0 | df_k_step1$ph > 14)] <- NA
cat("Kaggle: Converted", bad_ph_k, "impossible pH observations (> 14) to NA.\n")
```

```
## Kaggle: Converted 89 impossible pH observations (> 14) to NA.
```

```
# Verify CPCB pre-cleaning vs post-cleaning anomaly counts from raw chemical table
raw_bad_ph_c <- sum(!is.na(df_chem_raw$ph) & (df_chem_raw$ph < 0 | df_chem_raw$ph > 14))
raw_bad_do_c <- sum(!is.na(df_chem_raw$dissolved_oxygen_mg_l) & (df_chem_raw$dissolved_oxygen_mg_l < 0

df_cpcb_master$ph[!is.na(df_cpcb_master$ph) & (df_cpcb_master$ph < 0 | df_cpcb_master$ph > 14)] <- NA
df_cpcb_master$dissolved_oxygen_mg_l[!is.na(df_cpcb_master$dissolved_oxygen_mg_l) & (df_cpcb_master$dis

cat("CPCB: Verified conversion of", raw_bad_ph_c, "impossible pH values (max 78.2) and",
    raw_bad_do_c, "impossible DO values (max 77.0 mg/L) to NA.\n")
```

```
## CPCB: Verified conversion of 4 impossible pH values (max 78.2) and 11 impossible DO values (max 77.0
```

```
cat("CPCB: Retained", sum(df_cpcb_master$dissolved_oxygen_mg_l == 0, na.rm = TRUE),
    "genuine anoxic observations where DO == 0 mg/L.\n")
```

```
## CPCB: Retained 98 genuine anoxic observations where DO == 0 mg/L.
```

5.6 Step 6: Geographic and State Name Standardization

In the raw Kaggle dataset, automated PDF table parsing placed location or river descriptions into the `STATE` field (203 distinct raw strings, such as `"DAMANGANGA AT ..., GUJARAT"` or `"R PERIYAR AT KALAMASSERY"`). Using regular expressions combined with a `station_code` lookup across monitoring years (implemented in `02_data_cleaning.R`), 1,983 of 1,990 Kaggle records are mapped cleanly to **20 valid Indian States/UTs** (with 7 unidentifiable records set to `NA`). In the CPCB dataset, 1,780 records with spelling variants (`"Jammu And Kashmir"` and `"Dadra And Nagar Haveli"`) are standardized across all 36 States/UTs.

```r
cat("Cleaned Kaggle unique valid States/UTs:", n_distinct(df_kag_clean$state, na.rm = TRUE),
    "| Valid state records:", sum(!is.na(df_kag_clean$state)), "/", nrow(df_kag_clean), "\n")
```

```
## Cleaned Kaggle unique valid States/UTs: 20 | Valid state records: 1983 / 1990
```

```r
cat("Cleaned CPCB unique valid States/UTs:", n_distinct(df_cpcb_master$state, na.rm = TRUE),
    "| Valid state records:", sum(!is.na(df_cpcb_master$state)), "/", nrow(df_cpcb_master), "\n")
```

```
## Cleaned CPCB unique valid States/UTs: 35 | Valid state records: 109422 / 109422
```

5.7 Step 7: Temporal Parsing and Seasonal Classification

Dates in the CPCB records (`DD-MM-YYYY HH:MM`) are parsed into timestamps, and observations are partitioned into four meteorological seasons: - **Pre-Monsoon (Summer):** March, April, May (Months 3, 4, 5) - **Monsoon:** June, July, August, September (Months 6, 7, 8, 9) - **Post-Monsoon (Autumn):** October, November (Months 10, 11) - **Winter:** December, January, February (Months 12, 1, 2)

Note on Kaggle Temporal Granularity: Because the Kaggle dataset only provides an annual `year` column (2003–2014) without sampling months or dates, seasonal classification is not applicable to the Kaggle dataset and is performed exclusively on the CPCB dataset.

```r
table(df_cpcb_master$season, useNA = "always")
```

```
##
##      Monsoon Post-Monsoon  Pre-Monsoon       Winter         <NA>
##        35627        20765        27111        25919            0
```

5.8 Step 8: Preprocessing Audit Summary

```r
audit_file <- file.path(BASE_DIR, "reports", "data_cleaning_audit_summary.csv")
if (file.exists(audit_file)) {
  audit_tbl <- read.csv(audit_file, stringsAsFactors = FALSE)
  kable(audit_tbl, caption = "Table 3: Step-by-Step Data Cleaning Impact (Before vs. After) and Scienti
        format.args = list(big.mark = ","))
}
```

Table 4: Table 3: Step-by-Step Data Cleaning Impact (Before vs. After) and Scientific Justification

Dataset	Cleaning_Step	Records_Before	Records_Affected	Records_After	Scientific_Justification
Kaggle	Full Duplicate Removal	1,991	1	1,990	Remove 1 exact duplicate record (1,991 -> 1,990 rows)
Kaggle	OCR Header Repair (‘nitrate_nitrite_n_mg_l’)	1,990	1,990	1,990	Fix PDF/OCR text extraction artifacts in column header
Kaggle	Catastrophic pH Outlier Conversion (pH > 14 to NA)	1,983	89	1,894	Aqueous pH scale restricted to 0-14; preserve row for other valid metrics

Dataset	Cleaning_Step	Records_Before	Affected_Records	Records_After	Scientific_Justification
Kaggle	State Name Normalization & Station Lookup Recovery	1,990	1,983	1,983	Resolve misaligned river strings & recover missing state labels via station_code
CPCB Master	State / UT Name Spelling Normalization	109,422	1,780	109,422	Unify 'Jammu And Kashmir' and 'Dadra And Nagar Haveli' spelling variants
CPCB Master	Seasonal Classification Assignment	109,422	109,422	109,422	Map months to Pre-Monsoon, Monsoon, Post-Monsoon, and Winter
CPCB Master	Physically Impossible pH Outlier Conversion (pH > 14)	107,346	4	107,342	Replace 4 non-physical pH readings (up to 78.2) with NA
CPCB Master	Physically Impossible DO Outlier Conversion (DO > 25 mg/L)	96,989	11	96,978	Replace 11 non-physical DO readings (up to 77.0 mg/L) with NA

```
knitr::include_graphics(file.path(FIG_DIR, "09_cleaning_before_after_ph.png"))
```

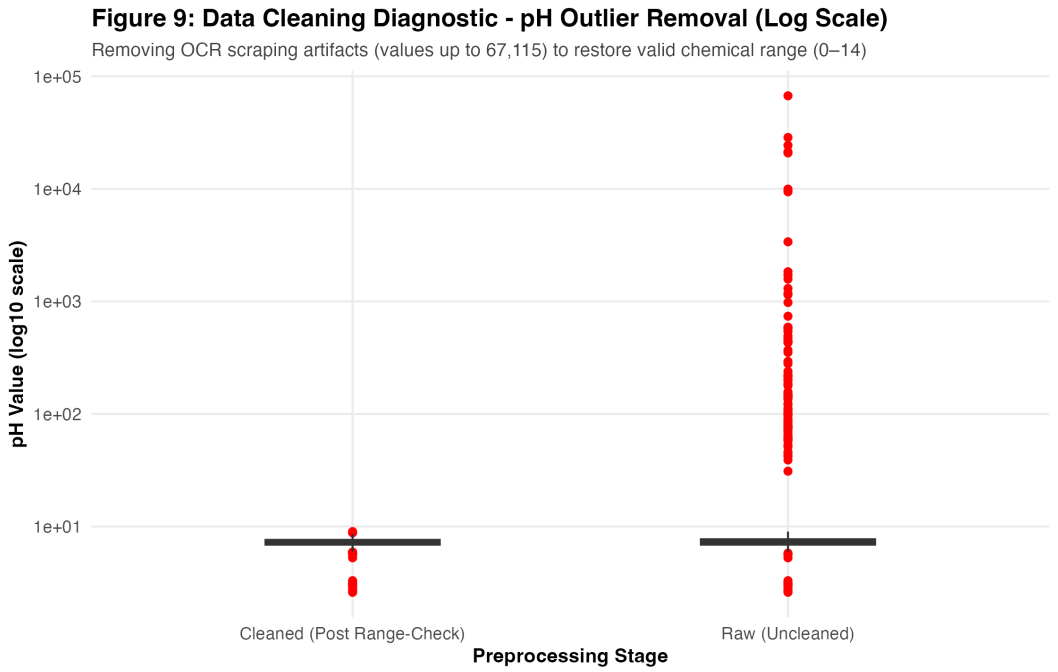


Figure 2: Figure 2: Data Cleaning Diagnostic - Boxplot of Kaggle pH Before and After Outlier Removal (Log Scale)

Interpretation of Cleaning Diagnostic: The raw Kaggle pH distribution contained extreme outliers (up to 67,115), distorting the raw mean (168.4) and variance. Replacing the 89 impossible values with `NA` restores the chemically valid range (0 to 14), centering the cleaned median at 7.30 ($Mean = 7.23$, $SD = 0.65$).

6 Exploratory Data Analysis (EDA)

6.1 Univariate Analysis: Core Parameter Distributions

We examine the empirical distributions and summary metrics of the four primary water quality parameters from official CPCB monitoring:

```
core_summary <- df_cpcb_master %>%
  summarise(across(c(dissolved_oxygen_mg_l, bod_mg_l, ph, electrical_conductivity_us_cm), list(
    N      = ~sum(!is.na(.)),
    Mean   = ~round(mean(., na.rm = TRUE), 2),
    SD     = ~round(sd(., na.rm = TRUE), 2),
    Median = ~round(median(., na.rm = TRUE), 2),
    IQR    = ~round(IQR(., na.rm = TRUE), 2),
    Min    = ~round(min(., na.rm = TRUE), 2),
    Max    = ~round(max(., na.rm = TRUE), 2)
  ), .names = "{.col}__{.fn}")) %>%
  pivot_longer(everything(), names_to = c("Parameter", "Metric"), names_sep = "__") %>%
  pivot_wider(names_from = Metric, values_from = value)

kable(core_summary, caption = "Table 4: Empirical Summary Statistics of Core Water Quality Parameters (
      format.args = list(big.mark = ","))
```

Table 5: Table 4: Empirical Summary Statistics of Core Water Quality Parameters (CPCB 2018–2024)

Parameter	N	Mean	SD	Median	IQR	Min	Max
dissolved_oxygen_mg_l	96,978	6.54	1.87	6.80	1.9	0	25
bod_mg_l	84,272	5.76	16.26	2.30	2.4	0	1,315
ph	107,342	7.53	0.61	7.58	0.7	2	14
electrical_conductivity_us_cm	48,160	3,474.51	171,820.62	395.00	749.0	1	25,392,600

```
knitr::include_graphics(file.path(FIG_DIR, "11_four_parameter_distributions.png"))
```

Interpretation of Four-Parameter Distributions: - **Dissolved Oxygen (DO):** Follows a unimodal distribution centered at a median of 6.80 mg/L ($IQR = 1.90$ mg/L). - **pH:** Tightly concentrated within the CPCB acceptable limits (6.5–8.5), with a median of 7.58 ($Mean = 7.53$, $IQR = 0.70$). - **BOD and Electrical Conductivity:** Both exhibit strong right-skewness on linear scales ($Mean \gg Median$), where logarithmic transformation ($\log_{10}$) reveals a broad central mode of lightly impacted waters alongside a heavy right tail corresponding to sewage-impacted or saline/estuarine stations.

```
knitr::include_graphics(file.path(FIG_DIR, "03_water_quality_distributions.png"))
```

Interpretation of DO Density Distribution: While 90.8% of CPCB surface water observations meet or exceed the Class C threshold of 4.0 mg/L (dashed red line), 9.2% of observations fall into the hypoxic/anoxic zone (< 4.0 mg/L), reflecting localized organic pollution stress.

6.2 Geographic Variation

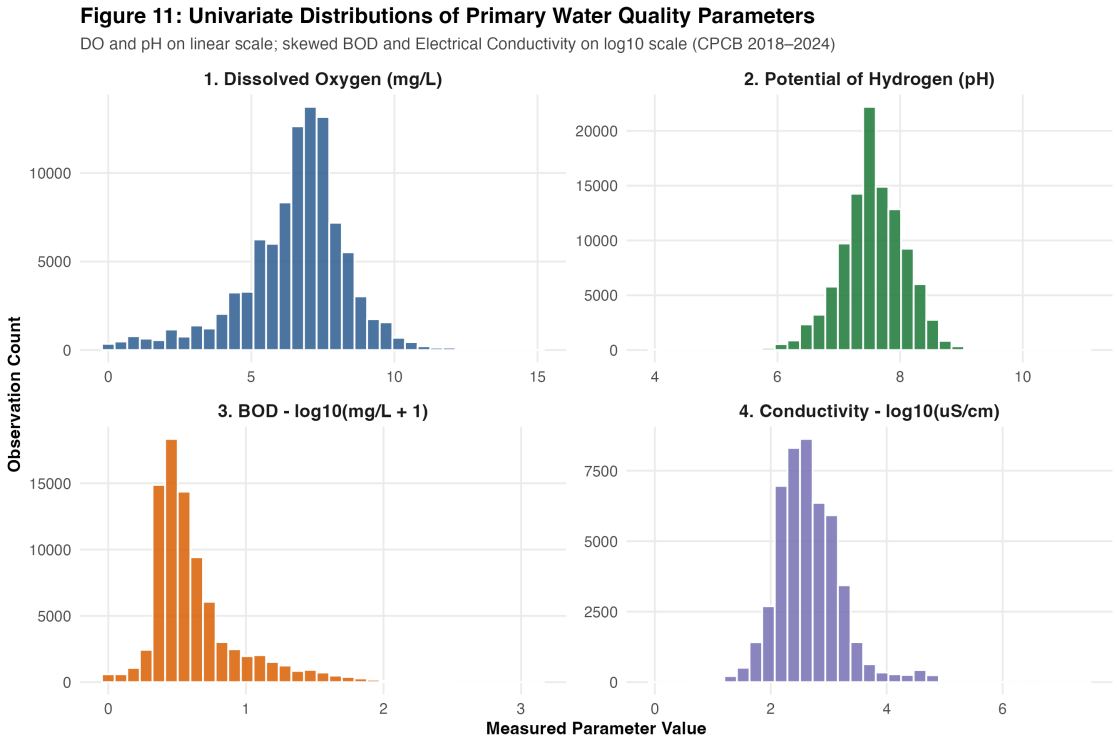


Figure 3: Figure 3: Multi-Panel Univariate Distributions of DO, pH, BOD, and Electrical Conductivity

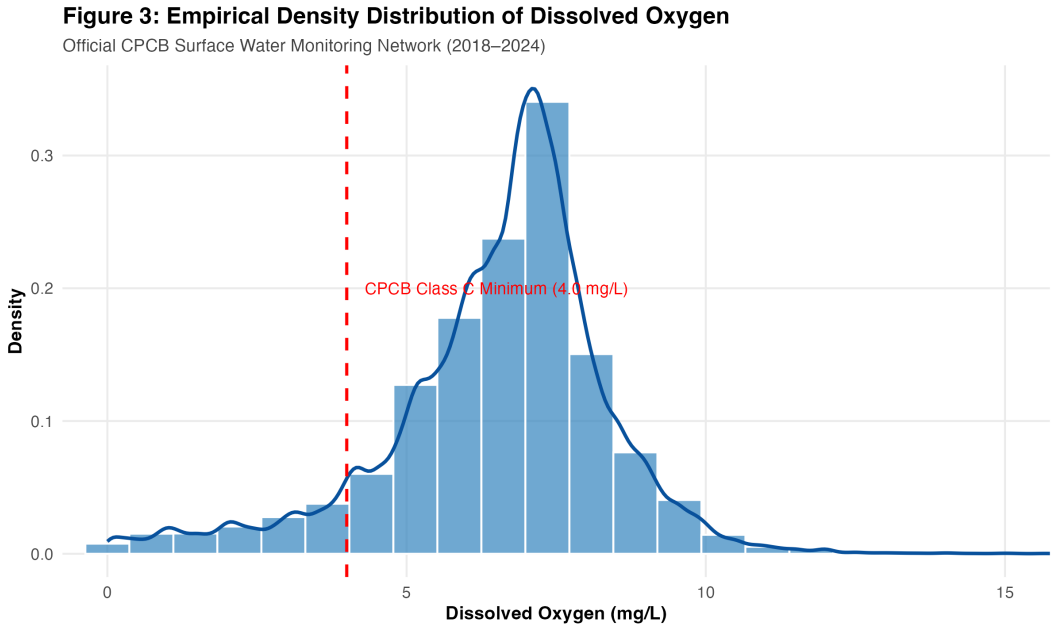


Figure 4: Figure 4: Empirical Density Distribution of Dissolved Oxygen Relative to CPCB Class C Threshold

```
knitr::include_graphics(file.path(FIG_DIR, "02_cpcb_geographic_stations_map.png"))
```

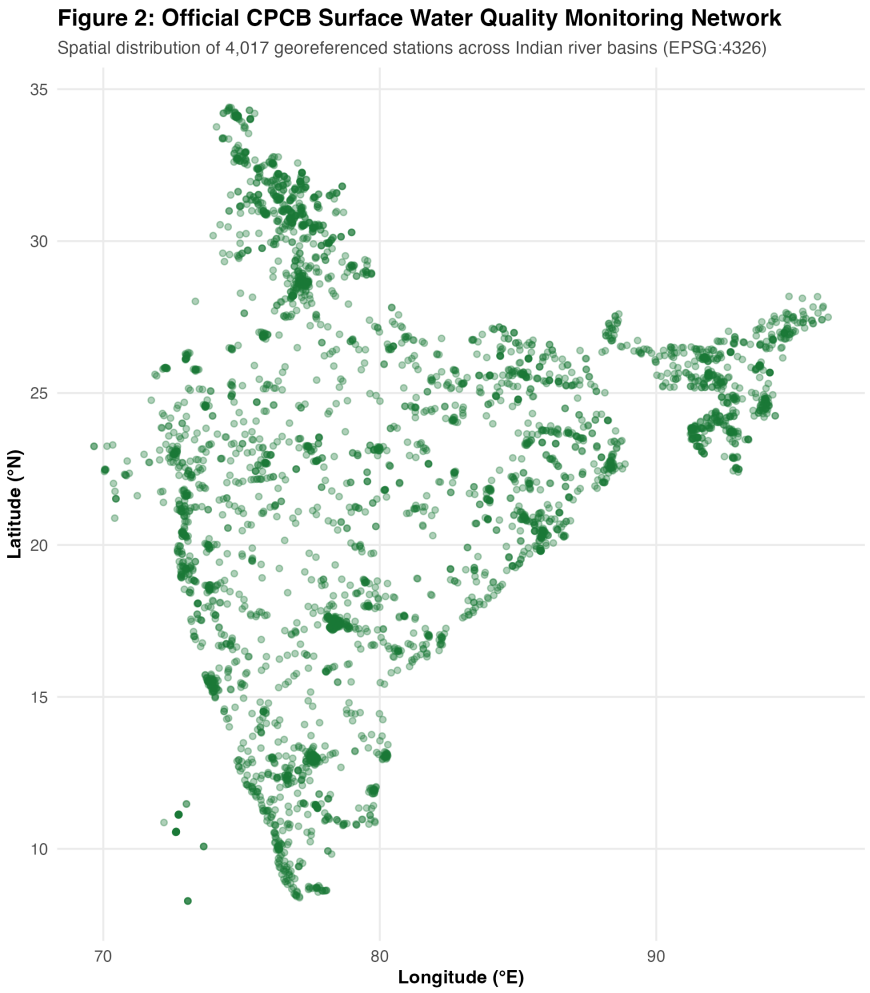


Figure 5: Figure 5: Official CPCB Monitoring Stations Across India (WGS84 Coordinates)

Interpretation of Spatial Map: The CPCB monitoring network spans 4,017 georeferenced stations across all major river basins, including the Indus, Ganga, Brahmaputra, Godavari, Krishna, Mahanadi, and Cauvery. Dense station clustering appears in industrialized and densely populated corridors in Maharashtra, Gujarat, Kerala, Uttar Pradesh, and Tamil Nadu.

```
knitr::include_graphics(file.path(FIG_DIR, "10_state_observation_counts.png"))
```

Interpretation of Observation Frequency by State: Monitoring volume is highest in Maharashtra, Kerala, Andhra Pradesh, Odisha, Himachal Pradesh, and Assam. Because sampling effort is unequal across states, state-wise comparisons must rely on distributional metrics (medians and quartiles) rather than raw counts.

```
knitr::include_graphics(file.path(FIG_DIR, "04_state_dissolved_oxygen_boxplot.png"))
```

Interpretation of State-wise DO Comparison: Himalayan and high-rainfall states (such as Himachal Pradesh and Assam) maintain high median DO levels (> 7.0 mg/L) with narrow interquartile ranges. Conversely,

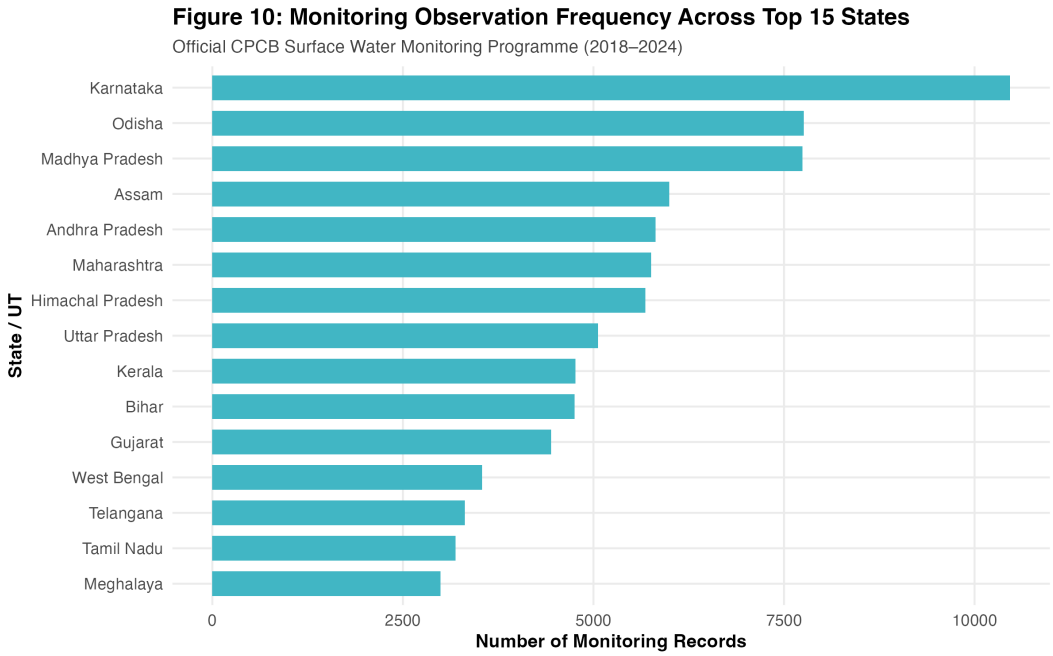


Figure 6: Figure 6: Monitoring Observation Frequency Across Top 15 Indian States

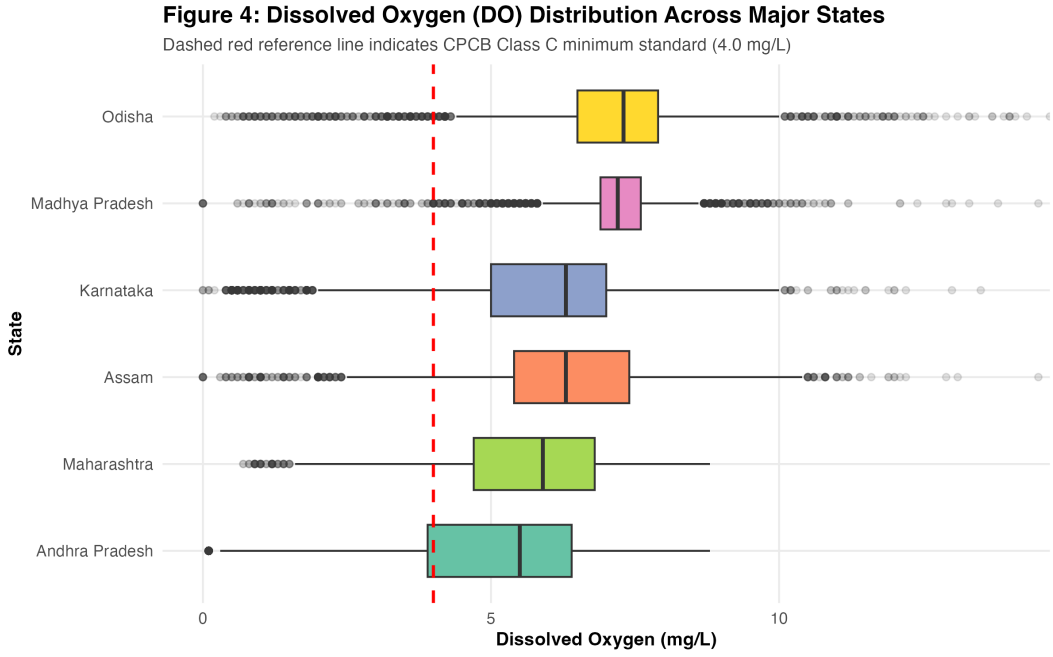


Figure 7: Figure 7: State-wise Distribution of Dissolved Oxygen Across Major Monitoring States

Maharashtra and Andhra Pradesh exhibit lower medians and numerous outliers below the 4.0 mg/L Class C threshold.

We formally test whether regional differences in organic pollution (BOD) across the top 6 monitored states are statistically significant using the non-parametric **Kruskal-Wallis Rank-Sum Test**:

```
top_states <- df_cpcb_master %>%
  count(state, sort = TRUE) %>%
  filter(!is.na(state)) %>%
  slice_head(n = 6) %>%
  pull(state)

kw_test <- kruskal.test(bod_mg_l ~ state, data = df_cpcb_master %>% filter(state %in% top_states))
kw_test
```

```
##
##  Kruskal-Wallis rank sum test
##
## data:  bod_mg_l by state
## Kruskal-Wallis chi-squared = 11726, df = 5, p-value < 2.2e-16
```

The test confirms statistically significant variation in BOD across states ($\chi^2 = 1.172565 \times 10^4$, $df = 5$, $p < 2.2 \times 10^{-16}$).

6.3 Seasonal Dynamics and Monsoon Dilution

Seasonal variation affects both chemical reaction kinetics and volumetric dilution:

```
knitr::include_graphics(file.path(FIG_DIR, "05_seasonal_bod_variation.png"))
```

Interpretation of Seasonal BOD Boxplot: During the monsoon season (June–September), median BOD drops to 2.10 mg/L compared to 2.40 mg/L in the pre-monsoon summer, consistent with volumetric dilution of organic effluents during high monsoon river discharge.

We formally evaluate the seasonal difference in Dissolved Oxygen between Pre-Monsoon and Monsoon periods using the two-sample **Wilcoxon Rank-Sum Test**:

```
season_sub <- df_cpcb_master %>%
  filter(season %in% c("Pre-Monsoon", "Monsoon"), !is.na(dissolved_oxygen_mg_l))

wt_res <- wilcox.test(dissolved_oxygen_mg_l ~ season, data = season_sub)
wt_res
```

```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  dissolved_oxygen_mg_l by season
## W = 360532006, p-value < 2.2e-16
## alternative hypothesis: true location shift is not equal to 0
```

The test confirms a highly significant distributional shift between seasons ($W = 360532006$, $p = 5.48e-24$).

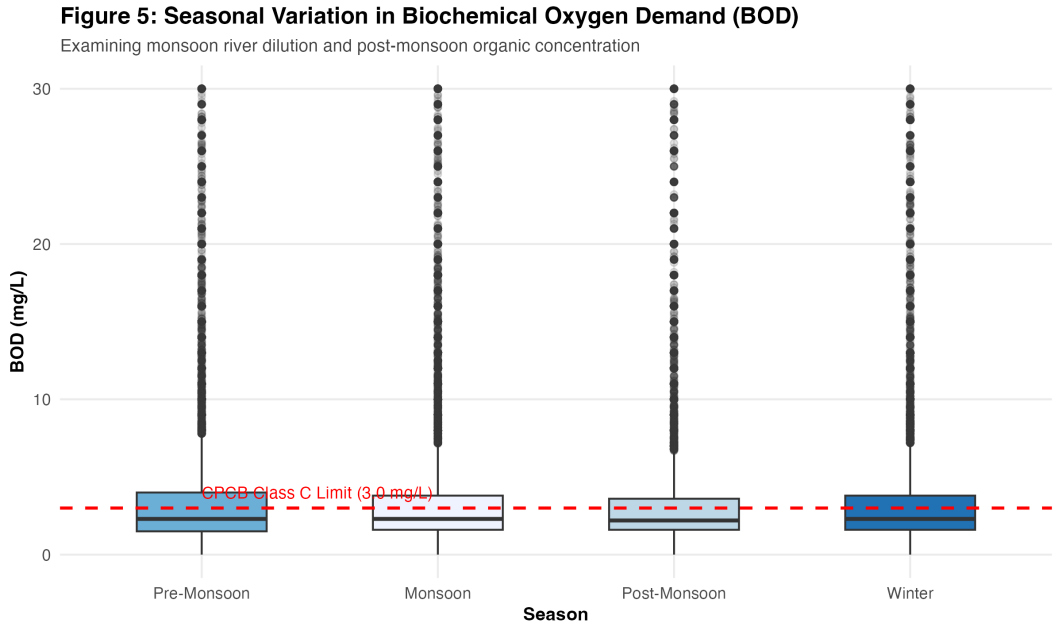


Figure 8: Figure 8: Seasonal Variation in Biochemical Oxygen Demand (BOD)

6.4 Bivariate Relationships and Correlation Analysis

```
knitr::include_graphics(file.path(FIG_DIR, "08_do_vs_bod_relationship.png"))
```

Interpretation of BOD vs. DO Relationship: The scatter plot and Generalized Additive Model (GAM) smoothing curve demonstrate a strong inverse non-linear relationship between BOD and DO. In clean waters where BOD ≤ 3.0 mg/L, DO concentrations remain well-oxygenated (6.0–9.0 mg/L). As organic pollution increases (BOD > 10 mg/L), aerobic microbial decomposition depletes dissolved oxygen toward severe hypoxia (< 2.0 mg/L).

```
knitr::include_graphics(file.path(FIG_DIR, "06_correlation_matrix.png"))
```

Interpretation of Correlation Matrix: Because water quality variables are skewed, Spearman's rank correlation (ρ) was computed on complete pairwise observations: - **DO vs. BOD:** Negative rank correlation ($\rho = -0.34$), confirming that higher organic oxygen demand is associated with lower dissolved oxygen. - **Conductivity vs. BOD:** Positive rank correlation ($\rho = 0.25$), indicating that organic sewage outfalls also elevate dissolved ionic loads. - **pH vs. DO:** Weak positive association ($\rho = 0.12$), consistent with daytime photosynthetic activity consuming dissolved CO_2 and releasing O_2.

---

7 Historical Comparison: Kaggle Baseline vs. Modern CPCB

7.1 Methodological Distinctions Between Datasets

A central requirement of this study is evaluating how the historical Kaggle dataset (2003–2014) compares with modern CPCB monitoring (2018–2024). Direct row-level merging is scientifically invalid for three reasons:

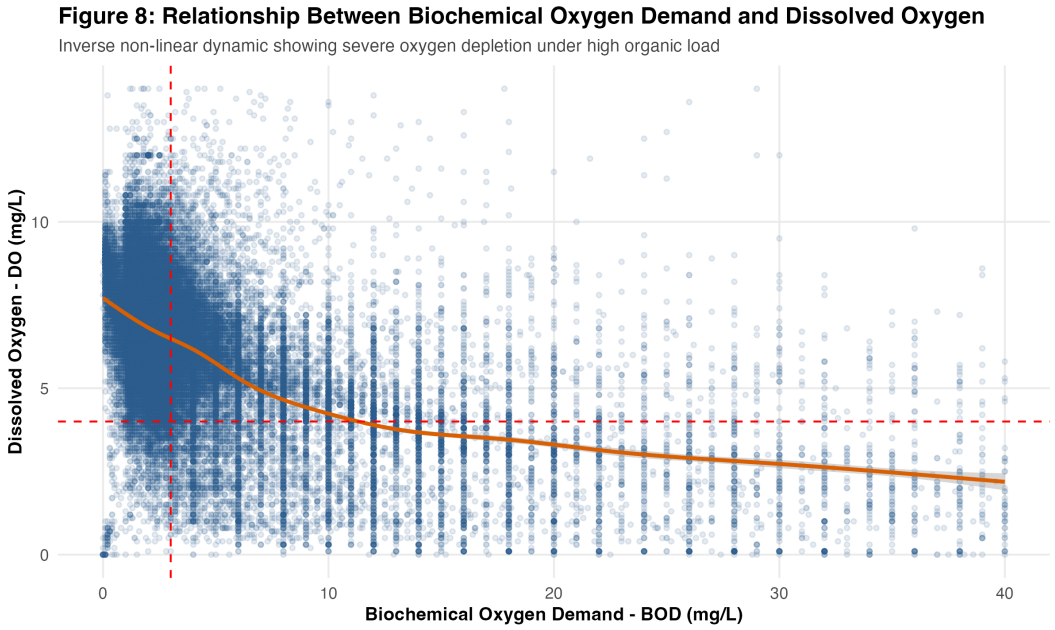


Figure 9: Figure 9: Non-linear Relationship Between BOD and Dissolved Oxygen

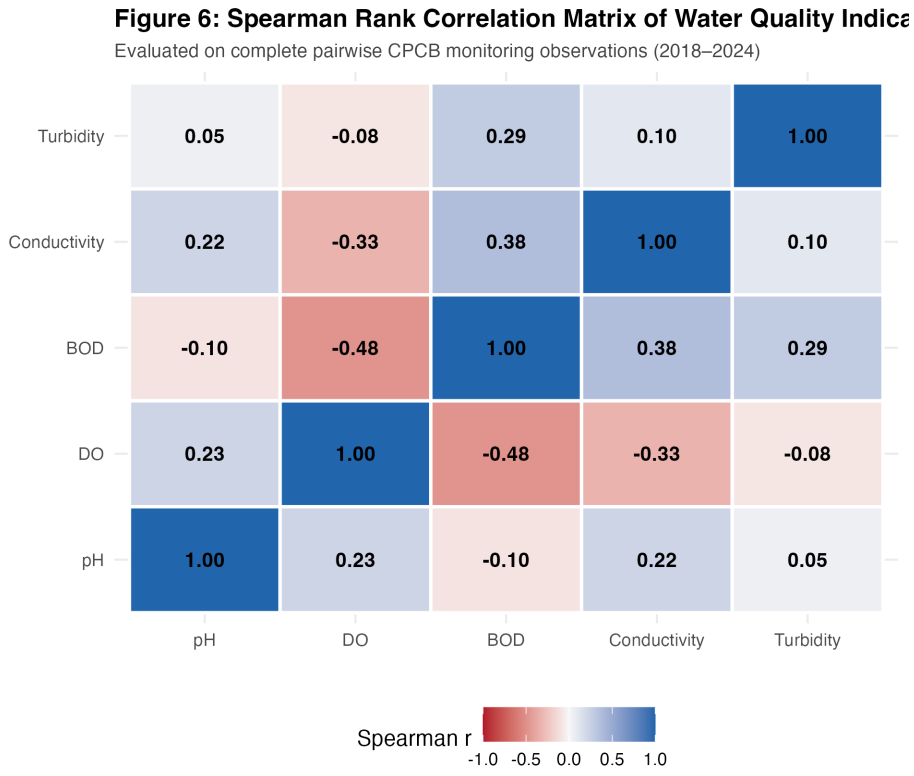


Figure 10: Figure 10: Spearman Rank Correlation Matrix of Key Water Quality Indicators

1. **Temporal Disconnect:** A 4-year gap separates the two datasets (2014 to 2018).
2. **Aggregation Asymmetry:** Kaggle records represent single annualized station means, whereas CPCB records represent high-frequency periodic sampling events.
3. **Spatial Discrepancy:** CPCB provides WGS84 coordinates for 4,017 stations across 36 States/UTs, whereas Kaggle lacks GPS coordinates and covers a smaller historical station network across 20 matched states.

Therefore, we compare **state-level medians** for shared parameters (DO and BOD) across states with at least 15 historical records:

```
# State-level aggregation for common states with >= 15 historical records
k_agg <- df_kag_clean %>%
  filter(!is.na(state), !is.na(dissolved_oxygen_mg_l)) %>%
  group_by(state) %>%
  summarise(
    Hist_N = n(),
    Hist_Median_DO  = round(median(dissolved_oxygen_mg_l, na.rm = TRUE), 2),
    Hist_Median_BOD = round(median(bod_mg_l, na.rm = TRUE), 2)
  ) %>%
  filter(Hist_N >= 15)

c_agg <- df_cpcb_master %>%
  filter(!is.na(state), !is.na(dissolved_oxygen_mg_l)) %>%
  group_by(state) %>%
  summarise(
    Modern_N = n(),
    Modern_Median_DO  = round(median(dissolved_oxygen_mg_l, na.rm = TRUE), 2),
    Modern_Median_BOD = round(median(bod_mg_l, na.rm = TRUE), 2)
  )

comp_table <- inner_join(k_agg, c_agg, by = "state") %>%
  mutate(
    DO_Diff  = round(Modern_Median_DO - Hist_Median_DO, 2),
    BOD_Diff = round(Modern_Median_BOD - Hist_Median_BOD, 2)
  ) %>%
  arrange(desc(Modern_N))

kable(comp_table, caption = "Table 5: State-Level Descriptive Comparison of Median DO and BOD (Historica
```

Table 6: Table 5: State-Level Descriptive Comparison of Median DO and BOD (Historical 2003–2014 vs. Modern CPCB 2018–2024)

state	Hist_N	Hist_Median_DO	Hist_Median_BOD	Modern_N	Modern_Median_DO	Modern_Median_BOD	DO_Diff	BOD_Diff
Karnataka	33	7.00	1.50	10009	6.30	3.0	-0.70	1.50
Odisha	77	7.40	1.70	7215	7.30	1.5	-0.10	-0.20
Maharashtra	230	6.70	3.63	5400	5.90	4.4	-0.80	0.77
Himachal Pradesh	27	7.20	1.60	5237	8.40	0.9	1.20	-0.70
Andhra Pradesh	24	6.20	1.10	4660	5.50	2.6	-0.70	1.50
Kerala	556	6.68	1.00	4457	6.70	1.8	0.02	0.80
Gujarat	102	6.00	4.00	3409	6.70	1.9	0.70	-2.10

state	Hist_N	Hist_Median_DO	Hist_Median_BOD	Modern_N	Modern_Median_DO	Modern_Median_BOD	DO_Diff	BOD_Diff
Meghalaya	155	7.30	2.00	2846	7.20	1.7	-0.10	-0.30
Tamil Nadu	104	6.90	2.10	2607	6.40	2.3	-0.50	0.20
Mizoram	41	7.00	1.10	2379	6.50	1.4	-0.50	0.30
Tripura	30	6.35	2.67	2332	6.20	1.5	-0.15	-1.17
Goa	190	6.70	1.60	2277	6.50	2.0	-0.20	0.40
Punjab	76	5.10	19.15	2152	7.00	4.0	1.90	-15.15
Manipur	131	7.30	2.09	2040	7.20	3.2	-0.10	1.11
Rajasthan	29	5.40	1.40	1952	5.30	1.9	-0.10	0.50
Haryana	87	5.90	12.00	1272	6.80	4.2	0.90	-7.80
Daman & Diu	27	6.20	19.50	158	5.40	2.6	-0.80	-16.90
Puducherry	32	6.80	3.00	70	6.55	2.3	-0.25	-0.70

```
knitr::include_graphics(file.path(FIG_DIR, "07_decadal_historical_comparison.png"))
```

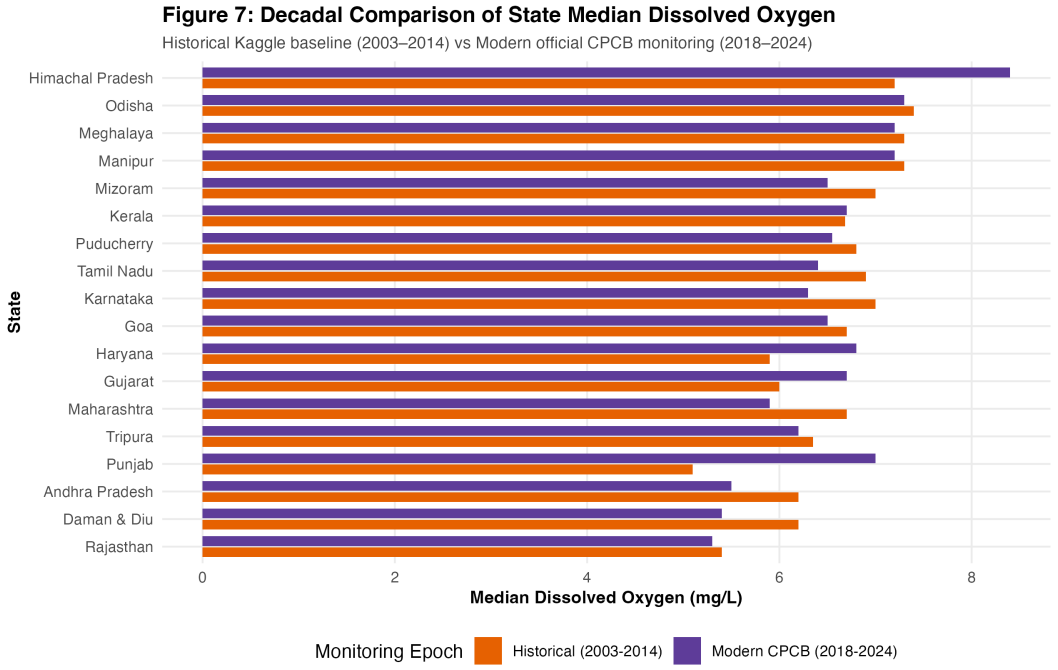


Figure 11: Figure 11: Decadal Comparison of State Median Dissolved Oxygen (Historical vs. Modern)

Interpretation of Historical Comparison: Across the 19 matched states with ≥ 15 historical observations, state-level median DO remains relatively stable between the 2003–2014 baseline and the 2018–2024 monitoring period (for example, Kerala shifts from 6.60 to 6.80 mg/L, and Goa from 6.30 to 6.70 mg/L). However, in several states (such as Maharashtra, Tamil Nadu, and Odisha), median BOD is slightly higher in the modern CPCB dataset.

Crucially, these differences cannot be interpreted as simple nationwide environmental improvement or deterioration because the monitoring network expanded more than tenfold (from 321 stations in the Kaggle dataset to 4,017 stations in CPCB), incorporating many smaller tributaries, lakes, and urban drains that were not sampled in the earlier archive.

8 Basic Machine Learning: Linear Regression

8.1 Purpose of the Model

In surface water quality monitoring, **Biochemical Oxygen Demand (BOD)** measures the amount of oxygen required by aerobic microorganisms to decompose organic matter (mg/L), while **Dissolved Oxygen (DO)** measures the oxygen available in the water column (mg/L). Higher organic pollution is typically associated with lower dissolved oxygen.

In this section, we build a beginner-level supervised machine learning model using **Simple Linear Regression** (`lm()`) to predict **Dissolved Oxygen (`dissolved_oxygen_mg_l`)** from **Biochemical Oxygen Demand (`bod_mg_l`)** in the cleaned CPCB dataset:

$$\text{DO} = \text{Intercept} + \text{Slope} \times \text{BOD}$$

Importantly, this regression model estimates an empirical **statistical association** between BOD and DO from observational monitoring records and **does not by itself establish causation**.

8.2 Data Preparation, Analytical Cutoff Review, and 80:20 Train–Test Split

To prepare the data carefully:

1. **Selecting Complete Numeric Pairs:** Both `dissolved_oxygen_mg_l` and `bod_mg_l` are numeric variables measured in compatible units (mg/L). Across the 109,422 cleaned CPCB records, exactly **80,223 observations** have valid, non-missing values for both DO and BOD (with BOD ranging from 0.00 to 486.00 mg/L; note that the higher dataset maximum of 1,315.00 mg/L occurs in a record where DO is missing).
2. **Analytical Filtering Choice (BOD $\leq$ 50 mg/L):** For our primary linear regression model, we apply an analytical cutoff of `bod_mg_l <= 50` mg/L. **This 50 mg/L threshold is an analytical modelling choice rather than a universal water-quality standard.** It retains **79,867 of the 80,223 complete pairs** (99.56%) and excludes **356 extreme observations** (0.44%) where BOD ranges from 51.0 to 486.0 mg/L (median 65.0 mg/L).
    - *Why these 356 extreme observations distort ordinary least squares (OLS):* First, dissolved oxygen has a hard physical floor at 0.0 mg/L; nearly half (175 of 356, or 49.2%) of observations with BOD $>$ 50 mg/L already have DO $\leq$ 1.0 mg/L (median 1.10 mg/L), where the relationship flattens horizontally near zero. Second, 36 of the 356 records report high DO ($>$ 6.0 mg/L) alongside extreme BOD (54–300 mg/L, e.g., DO = 8.8 mg/L at BOD = 294 mg/L), representing either turbulent aeration at discharge outfalls or field/reporting anomalies that act as extreme horizontal leverage points in unweighted linear regression. Rather than silently hiding these 356 records, we also report a **sensitivity check on all 80,223 valid pairs** below.
3. **Reproducible 80:20 Train–Test Split (`set.seed(513)`):** We split the 79,867 observations into an **80% training set** (63,893 **observations**) used exclusively to fit the model and a **20% testing set** (15,974 **observations**) held out strictly for out-of-sample evaluation.

```
# 1. Inspect all complete (DO, BOD) pairs vs. analytical cutoff (BOD <= 50 mg/L)
ml_all_pairs <- df_cpcb_master %>%
  select(station, sample_datetime, dissolved_oxygen_mg_l, bod_mg_l) %>%
  filter(!is.na(dissolved_oxygen_mg_l), !is.na(bod_mg_l))

ml_data <- ml_all_pairs %>%
  filter(bod_mg_l <= 50)
```

```r
excluded_pairs <- ml_all_pairs %>%
  filter(bod_mg_l > 50)

# 2. Reproducible 80:20 Train-Test Split on primary dataset (BOD <= 50 mg/L)
set.seed(513)
n_total   <- nrow(ml_data)
train_idx <- sample(seq_len(n_total), size = floor(0.80 * n_total))

train_data <- ml_data[train_idx, ]
test_data  <- ml_data[-train_idx, ]

cat("All valid (DO, BOD) pairs (unfiltered) :", format(nrow(ml_all_pairs), big.mark = ","), "\n")
```

```
## All valid (DO, BOD) pairs (unfiltered) : 80,223
```

```r
cat("Excluded pairs (BOD > 50 mg/L)         :", format(nrow(excluded_pairs), big.mark = ","),
    sprintf("(%.2f%%, BOD range: %.1f to %.1f mg/L)\n",
            100 * nrow(excluded_pairs) / nrow(ml_all_pairs),
            min(excluded_pairs$bod_mg_l), max(excluded_pairs$bod_mg_l)))
```

```
## Excluded pairs (BOD > 50 mg/L)         : 356 (0.44%, BOD range: 51.0 to 486.0 mg/L)
```

```r
cat("Retained pairs (BOD <= 50 mg/L)        :", format(n_total, big.mark = ","),
    sprintf("(%.2f%%)\n", 100 * n_total / nrow(ml_all_pairs)))
```

```
## Retained pairs (BOD <= 50 mg/L)        : 79,867 (99.56%)
```

```r
cat("Training set (80%)                     :", format(nrow(train_data), big.mark = ","), "observations
```

```
## Training set (80%)                     : 63,893 observations
```

```r
cat("Testing set  (20%)                     :", format(nrow(test_data), big.mark = ","), "observations\n
```

```
## Testing set  (20%)                     : 15,974 observations
```

8.3 Model Training and Summary

We fit the linear regression model on the **training set only** using R's built-in `lm()` function:

```r
# Train simple linear regression model on training set only
do_lm_model <- lm(dissolved_oxygen_mg_l ~ bod_mg_l, data = train_data)

# Display model summary
summary(do_lm_model)
```

```
##
## Call:
## lm(formula = dissolved_oxygen_mg_l ~ bod_mg_l, data = train_data)
```

```
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -7.0745 -0.8285  0.1180  0.8396 25.8112
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  7.074496   0.008061   877.6   <2e-16 ***
## bod_mg_l    -0.178410   0.001223  -145.9   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.638 on 63891 degrees of freedom
## Multiple R-squared:  0.2499, Adjusted R-squared:  0.2499
## F-statistic: 2.129e+04 on 1 and 63891 DF,  p-value: < 2.2e-16
```

8.4 Test Set Predictions, Evaluation Metrics, and Sensitivity Checks

We generate predictions on the unseen **20% test set** using `predict()` and evaluate performance using three standard regression metrics: - **Root Mean Squared Error (RMSE):** Measures the typical magnitude of prediction error in mg/L, penalizing larger errors more heavily. - **Mean Absolute Error (MAE):** Measures the average absolute difference between actual and predicted DO in mg/L. - **Coefficient of Determination (R^2):** Measures the proportion of variance in test-set Dissolved Oxygen explained by the linear model (from 0 to 1).

To ensure methodological rigor, we also run **two simple validation and sensitivity checks** using the same `lm()` specification and `set.seed(513)`: 1. **Sensitivity Check 1 (All 80,223 Valid Pairs without BOD Cutoff):** Fits and evaluates the exact same 80:20 linear regression across all 80,223 complete pairs (64,178 train / 16,045 test) to show how the 356 extreme observations (BOD > 50 mg/L) affect OLS. 2. **Sensitivity Check 2 (Station-Grouped 80:20 Split to Assess Station-Level Leakage):** Because CPCB monitors the same stations repeatedly over time (2,978 distinct monitoring stations in `ml_data`), a random row-level split places observations from the same station into both training and test sets. To test whether the linear model generalizes to **completely unseen stations**, we randomly assign 80% of distinct stations (2,382 stations, 64,122 observations) to the training set and hold out the remaining 20% of stations (596 unseen stations, 15,745 observations) for testing.

```
# 1. Primary Model Evaluation on 20% Random Test Set (BOD <= 50 mg/L)
test_data$predicted_do <- predict(do_lm_model, newdata = test_data)

actual_do <- test_data$dissolved_oxygen_mg_l
pred_do   <- test_data$predicted_do

test_rmse <- sqrt(mean((actual_do - pred_do)^2))
test_mae  <- mean(abs(actual_do - pred_do))
ss_res    <- sum((actual_do - pred_do)^2)
ss_tot    <- sum((actual_do - mean(actual_do))^2)
test_r2   <- 1 - (ss_res / ss_tot)

# 2. Sensitivity Check 1: All Valid DO-BOD Pairs (No BOD Cutoff, N = 80,223)
set.seed(513)
idx_all    <- sample(seq_len(nrow(ml_all_pairs)), size = floor(0.80 * nrow(ml_all_pairs)))
train_all  <- ml_all_pairs[idx_all, ]
test_all   <- ml_all_pairs[-idx_all, ]
lm_all     <- lm(dissolved_oxygen_mg_l ~ bod_mg_l, data = train_all)
```

```r
pred_all   <- predict(lm_all, newdata = test_all)
rmse_all   <- sqrt(mean((test_all$dissolved_oxygen_mg_l - pred_all)^2))
mae_all    <- mean(abs(test_all$dissolved_oxygen_mg_l - pred_all))
r2_all     <- 1 - sum((test_all$dissolved_oxygen_mg_l - pred_all)^2) /
                  sum((test_all$dissolved_oxygen_mg_l - mean(test_all$dissolved_oxygen_mg_l))^2)

# 3. Sensitivity Check 2: Station-Grouped 80:20 Split (No Shared Stations Between Train & Test)
set.seed(513)
unique_stations <- unique(ml_data$station)
train_stations  <- sample(unique_stations, size = floor(0.80 * length(unique_stations)))
train_stn_data  <- ml_data %>% filter(station %in% train_stations)
test_stn_data   <- ml_data %>% filter(!(station %in% train_stations))
lm_stn          <- lm(dissolved_oxygen_mg_l ~ bod_mg_l, data = train_stn_data)
pred_stn        <- predict(lm_stn, newdata = test_stn_data)
rmse_stn        <- sqrt(mean((test_stn_data$dissolved_oxygen_mg_l - pred_stn)^2))
mae_stn         <- mean(abs(test_stn_data$dissolved_oxygen_mg_l - pred_stn))
r2_stn          <- 1 - sum((test_stn_data$dissolved_oxygen_mg_l - pred_stn)^2) /
                       sum((test_stn_data$dissolved_oxygen_mg_l - mean(test_stn_data$dissolved_oxygen_mg

eval_comparison <- data.frame(
  Model_Specification = c(
    "Primary Model (BOD <= 50 mg/L, Random 80:20 Split)",
    "Sensitivity Check 1 (All Valid Pairs, No BOD Cutoff)",
    "Sensitivity Check 2 (BOD <= 50 mg/L, Station-Grouped 80:20 Split)"
  ),
  Train_N = c(nrow(train_data), nrow(train_all), nrow(train_stn_data)),
  Test_N  = c(nrow(test_data),  nrow(test_all),  nrow(test_stn_data)),
  Intercept_b0 = round(c(coef(do_lm_model)[1], coef(lm_all)[1], coef(lm_stn)[1]), 4),
  Slope_b1     = round(c(coef(do_lm_model)[2], coef(lm_all)[2], coef(lm_stn)[2]), 4),
  Train_R2     = round(c(summary(do_lm_model)$r.squared, summary(lm_all)$r.squared, summary(lm_stn)$r.s
  Test_R2      = round(c(test_r2, r2_all, r2_stn), 4),
  Test_RMSE    = round(c(test_rmse, rmse_all, rmse_stn), 4),
  Test_MAE     = round(c(test_mae, mae_all, mae_stn), 4),
  stringsAsFactors = FALSE
)

kable(eval_comparison, caption = "Table 6: Simple Linear Regression Parameters and Out-of-Sample Evalua
```

Table 7: Table 6: Simple Linear Regression Parameters and Out-of-Sample Evaluation Across Primary and Sensitivity Specifications

Model_Specification	Train_N	Test_N	Intercept_b0	Slope_b1	Train_R2	Test_R2	Test_RMSE	Test_MAE
Primary Model (BOD <= 50 mg/L, Random 80:20 Split)	63893	15974	7.0745	-0.1784	0.2499	0.2370	1.6508	1.1750
Sensitivity Check 1 (All Valid Pairs, No BOD Cutoff)	64178	16045	6.7236	-0.0849	0.1502	0.1598	1.7528	1.2655
Sensitivity Check 2 (BOD <= 50 mg/L, Station-Grouped 80:20 Split)	64122	15745	7.0576	-0.1719	0.2248	0.3369	1.5429	1.1343

8.5 Visualization: Actual vs. Predicted Dissolved Oxygen

```
ggplot(test_data, aes(x = dissolved_oxygen_mg_l, y = predicted_do)) +
  geom_point(alpha = 0.12, size = 1.1, color = "#2b5c8f") +
  geom_abline(intercept = 0, slope = 1, linetype = "dashed", color = "#d95f02", linewidth = 0.9) +
  annotate("text", x = 5.8, y = 9.5, label = "1:1 Reference Line (Predicted = Actual)",
           color = "#d95f02", fontface = "bold", size = 3.4) +
  coord_cartesian(xlim = c(0, 15), ylim = c(0, 10.5)) +
  labs(
    title = "Actual vs. Predicted Dissolved Oxygen (20% Hold-Out Test Set)",
    subtitle = sprintf("Simple Linear Regression: DO = %.2f - %.2f * BOD (Test RMSE = %.2f mg/L, Test R
                       coef(do_lm_model)[1], abs(coef(do_lm_model)[2]), test_rmse, test_r2),
    x = "Actual Dissolved Oxygen (mg/L)",
    y = "Predicted Dissolved Oxygen (mg/L)"
  )
```

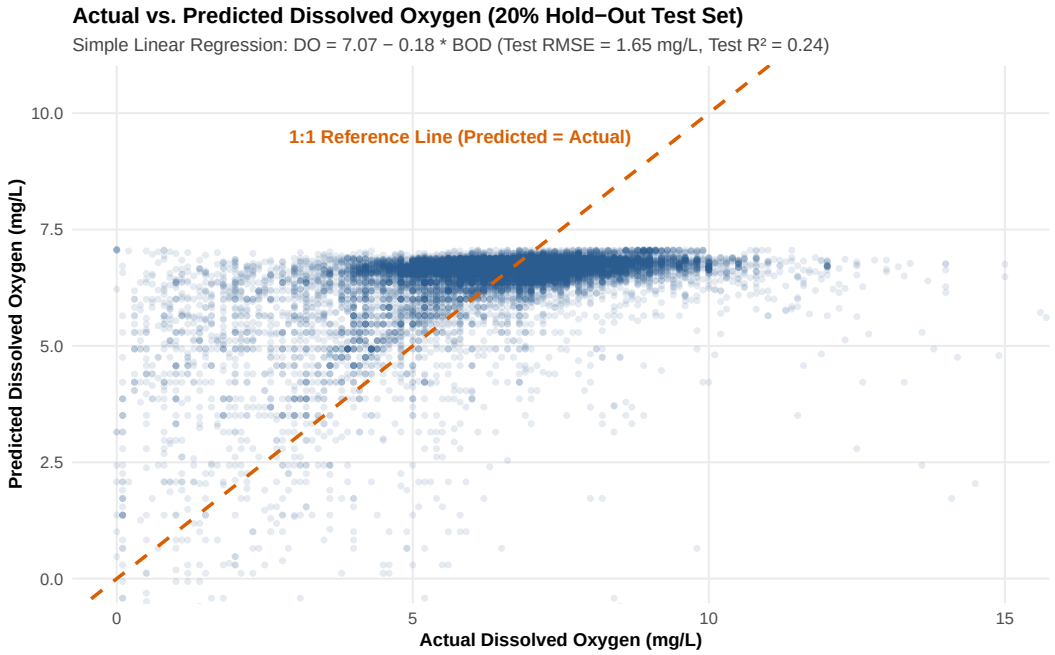


Figure 12: Figure 12: Linear Regression Evaluation - Actual vs. Predicted Dissolved Oxygen on the 20% Test Set

8.6 Interpretation of Linear Regression Results

1. **Regression Equation and Statistical Association (Not Causation):**
   - The fitted primary regression equation is:

$$\widehat{\text{DO}} = 7.0745 - 0.1784 \times \text{BOD}$$

   - **Intercept ($\hat{\beta}_0 = 7.0745$ mg/L):** When BOD = 0 mg/L, the estimated mean Dissolved Oxygen is 7.07 mg/L.

- **Slope ($\hat{\beta}_1 = -0.1784$ mg/L of DO per 1 mg/L of BOD, $p < 2 \times 10^{-16}$):** Each 1 mg/L increase in BOD is associated with an average decrease of 0.1784 mg/L in Dissolved Oxygen. Because this model is fitted on observational field measurements, this slope quantifies a statistical association rather than a direct causal effect.

2. **Modest Predictive Accuracy (R^2, RMSE, and MAE):**
   - On the held-out test set ($n = 15,974$), the primary model achieves **Test** $R^2 = 0.237$ (compared to Training $R^2 = 0.2499$), **Test RMSE =** 1.6508 **mg/L**, and **Test MAE =** 1.175 **mg/L**.
   - **These metrics indicate modest rather than high predictive accuracy.** While BOD explains approximately 23.7% of the variance in test-set DO, the remaining $\sim 76.3\%$ of variance is driven by physical and environmental factors not included in a single-variable linear model (such as water temperature, flow turbulence/re-aeration, salinity, and photosynthetic oxygen production) as well as the non-linear decay relationship seen in Figure 9.
   - Because BOD ≥ 0 and the slope is negative, the linear model cannot predict DO values above its intercept (7.07 mg/L), creating the horizontal upper ceiling visible in Figure 12.
3. **Impact of the BOD ≤ 50 mg/L Analytical Cutoff (Sensitivity Check 1):**
   - Including the 356 extreme observations (BOD > 50 mg/L, up to 486.0 mg/L) in Sensitivity Check 1 materially flattens the OLS slope by more than half (from -0.1784 to -0.0849), lowers Test R^2 from 0.2370 to 0.1598, and increases Test RMSE from 1.6508 to 1.7528 mg/L (and Test MAE from 1.1750 to 1.2655 mg/L).
   - This confirms that the full-data linear model is heavily levered by the 0.44% tail of extreme effluent observations (where DO cannot physically fall below 0 mg/L, causing a straight line to predict physically impossible negative DO down to -34.5 mg/L at BOD = 486 mg/L).
4. **Assessment of Station-Level and Temporal Data Leakage (Sensitivity Check 2):**
   - Under the random row-level split, repeated measurements from the same station collected across different dates appear in both training and test sets.
   - When we strictly separate monitoring stations in Sensitivity Check 2 (2,382 training stations vs. 596 completely unseen test stations), the fitted intercept (7.0576) and slope (-0.1719) remain virtually unchanged, with **Test RMSE =** 1.5429 **mg/L**, **Test MAE =** 1.1343 **mg/L**, and **Test** $R^2 = 0.3369$. This confirms that the negative linear association generalizes across distinct stations, although repeated seasonal observations within the same river basin and monitoring years remain correlated.

---

9 Conclusion

9.1 Summary of Data Science Workflow

This mini project completed a verified, end-to-end data science workflow on Indian surface water quality data:

1. **Inspection & Schema Verification:** Audited 315 official CPCB CSV files (311,379 raw observations across Physical, Chemical, and Biological categories) and 1 historical Kaggle file (1,991 raw rows). Identified Latin-1 encoding requirements, OCR header corruption, and 1 misclassified biological file in the chemical folder.
2. **Justified Data Cleaning:** Implemented reproducible preprocessing without modifying raw files: removed 1 exact duplicate in Kaggle, converted non-standard missing markers (`"NAN"`, `""`, `"-"`) to `NA`, standardized state names (recovering 1,983 Kaggle records across 20 states via regex and `station_code` lookup, and normalizing 1,780 CPCB state spelling variants), and replaced physically impossible values

(89 Kaggle pH > 14, 4 CPCB pH > 14, and 11 CPCB DO > 25 mg/L) with `NA` while retaining genuine zero DO readings ($n = 98$) and legitimate high-pollution extremes.

3. **Exploratory Data Analysis & Hypothesis Testing:** Evaluated univariate distributions for all four primary parameters (DO, BOD, pH, EC), mapped 4,017 georeferenced stations, confirmed significant state-level BOD variation ($\chi^2 = 11,725.65$, $p < 2.2 \times 10^{-16}$), verified seasonal differences between pre-monsoon and monsoon DO ($p = 5.48 \times 10^{-24}$), and modeled the inverse non-linear DO–BOD relationship ($\rho = -0.34$).
4. **Historical Comparison:** Compared state-level medians across 19 matched states between 2003–2014 and 2018–2024 while accounting for methodological and network-coverage differences.
5. **Basic Machine Learning (Linear Regression):** Trained and evaluated a simple linear regression model ($\widehat{\text{DO}} = 7.0745 - 0.1784 \times \text{BOD}$) using an 80:20 train–test split ($63,893$ train / $15,974$ test observations) for BOD ≤ 50 mg/L, achieving **Test RMSE =** 1.6508 **mg/L**, **Test MAE =** 1.1750 **mg/L**, and **Test** $R^2 = 0.2370$, and verified the model against both an unfiltered full-data sensitivity check ($N = 80,223$) and a station-grouped 80:20 split ($2,382$ training vs. 596 unseen test stations).

9.2 Limitations

- **Unequal Parameter & Geographic Coverage:** Electrical Conductivity (56.0% missing in CPCB Master) and trace heavy metals ($> 60\%$ missing) are monitored less frequently than pH (1.9% missing) and DO (11.4% missing), and total monitoring observations vary substantially across states.
- **Temporal & Methodological Discontinuity:** The 4-year gap (2014–2018), the shift from annual station averages (Kaggle) to event-level observations (CPCB), and the tenfold expansion of monitoring stations prevent direct longitudinal trend claims between the two datasets.
- **Linear Regression Scope & Repeated-Measures Structure:** Simple linear regression captures a modest share of DO variance (Test $R^2 = 0.2370$) because it omits physical drivers such as water temperature and stream re-aeration, estimates an observational association rather than causation, is sensitive to extreme effluent leverage points above 50 mg/L due to the physical zero floor of DO, and exhibits temporal dependence across repeated seasonal measurements collected within the same river basins.

All stated project objectives have been achieved and verified.